Commodity Forward Curve Analytics Platform

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1 Introduction

1.1 What is a Forward Curve?

When a commodity is traded on a futures exchange, it is not only the current price that is visible — dozens of contracts with different delivery dates are quoted simultaneously. The set of all these prices, plotted against their maturity, forms what is called a **forward curve**.

The shape of this curve is not random. It directly reflects the market's collective view of supply, demand, and storage conditions over the coming months. Two fundamental structures exist:

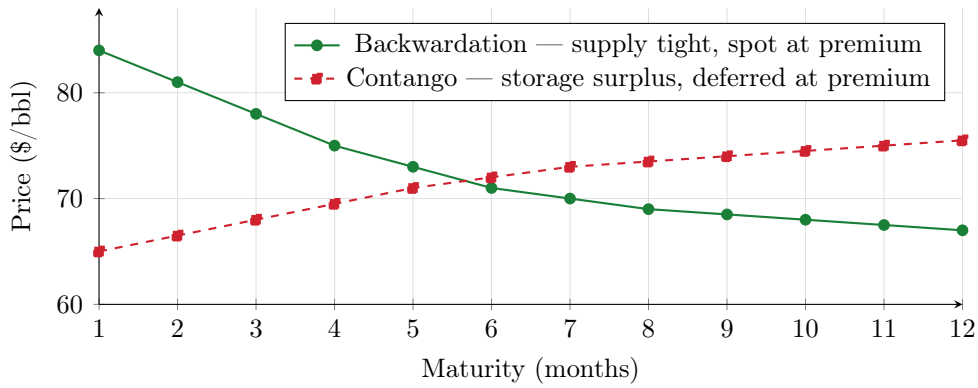

Figure 1: The two main forward curve structures.

Key definitions

Backwardation: near-term prices higher than deferred prices. The market is paying a premium for immediate delivery, signalling tight supply or low inventories.

Contango: deferred prices higher than near-term prices. Storage costs and a comfortable supply balance push future prices above today's spot.

1.2 Why Does It Matter?

Reading the forward curve correctly allows traders, risk managers and industrial buyers to answer practical questions every day:

- Is the market tight right now, or are inventories building?
- Should I lock in a fixed price today, or wait for deferred contracts?
- Does the market expect the current price level to persist, or to mean-revert?

- What is the implied cost of rolling a long futures position forward each month?

1.3 What is CFCAP?

CFCAP (Commodity Forward Curve Analytics Platform) is a Python application that automates the full analysis pipeline: it downloads live futures prices, fits a parametric model to the curve, computes financial metrics, generates trading signals, and presents all results in an interactive dashboard.

It covers more than **65 commodities** across 8 asset classes, from WTI Crude Oil to LME Nickel, EU Carbon allowances and Baltic freight rates.

2 Architecture and Usage

2.1 A Single File, Four Modes

The entire platform is contained in a single Python file, `cfcap.py`. It can be used in four different ways depending on the context:

Mode	Description	How to launch
Desktop dialog	Tkinter GUI — select commodity, set RF rate and horizon, click Run	`python cfcap.py`
Streamlit dashboard	Full browser interface with interactive Plotly charts	`streamlit run cfcap.py`
CLI single run	One commodity, automated, no interaction needed	`python cfcap.py --commodity X`
Daily scheduler	Batch of 11 commodities every morning at 09:15 EST	`python cfcap.py --schedule`

2.2 The Analysis Pipeline

Every run, regardless of mode, follows the same six-step pipeline:

2.3 Technologies and Libraries

Library	Role	Purpose in CFCAP
`numpy`	**Numerics**	Curve fitting, yield computations, VaR estimates
`pandas`	**Data**	Forward curve DataFrames, CSV snapshots, historical loads
`scipy`	**Optimisation**	`curve_fit` for Nelson-Siegel, cubic spline interpolation
`matplotlib`	**Visualisation**	Static 4-panel PNG dashboard
`yfinance`	**Market data**	Live futures prices (NYMEX, COMEX, CBOT, ICE)
`tvdatafeed`	**Market data**	TradingView contracts: LME, TTF, Jet Kero, Carbon
`streamlit`	**Dashboard**	Interactive browser interface
`plotly`	**Charts**	Zoomable, hoverable charts in Streamlit
`requests`	**API**	EIA Open Data: crude and gas inventories, production
`schedule`	**Automation**	Daily batch scheduler
`tkinter`	**GUI**	Desktop commodity selection dialog

3 Commodities Covered

The commodity registry covers 65 instruments. Each entry stores its data source, CME month codes, ticker format, storage cost, and Nelson-Siegel fitting bounds. The program routes each download automatically.

Family	Count	Source	Examples
Energy	9	Yahoo Finance	WTI, Brent, Natural Gas, RBOB, Heating Oil, Gasoil, Jet Kero
Metals	5	Yahoo Finance	Gold, Silver, Copper, Platinum, Palladium
Agriculture	15	Yahoo Finance	Corn, Wheat, Soybeans, Sugar, Coffee, Cocoa, Cattle, Hogs
Base Metals (LME)	10	TradingView	Copper, Aluminum, Zinc, Nickel, Lead, Tin, Cobalt, Lithium
Energy (additional)	10	TradingView	TTF, NBP, Coal API2/4, Uranium, Carbon EUA, Naphtha
Freight	4	TradingView	Capesize, Panamax, Supramax, VLCC Tanker
Carbon	4	TradingView	EU EUA, UK UKA, California CCA, RGGI
Dairy & Tropical	8	Yahoo / TV	Palm Oil, Canola, Milk, Butter, Robusta Coffee
Total	**65+**		

Data routing: Yahoo Finance is used for most exchange-traded contracts via a single grouped download. TradingView is used for ICE London, LME and other contracts that are not available on Yahoo. The routing is automatic and transparent to the user.

4 Forward Curve Analytics

4.1 The Nelson-Siegel Model

The Nelson-Siegel model (1987) was originally developed for fixed-income yield curves. Applied to commodity futures, it fits a smooth parametric function to the observed prices and extracts four parameters with direct economic meaning.

$$F(T) = \beta_0 + \beta_1 \cdot \varphi(T,\tau) + \beta_2 \cdot \left[\varphi(T,\tau) - e^{-T/\tau}\right], \quad \varphi(T,\tau) = \frac{1 - e^{-T/\tau}}{T/\tau} \tag{1}$$

Param.	**Name**	What it tells us	Example (WTI)
β_0	**Long-term level**	The price the curve converges to as maturity grows — the model's estimate of long-run fair value. If spot is well above β_0, the market expects prices to fall.	68 \$/bbl
β_1	**Slope**	The difference between the current spot and the long-term level. A large negative β_1 means strong backwardation — the front of the curve is much higher than the back.	−8 \$/bbl
β_2	**Curvature**	A hump or valley in the middle of the curve. This arises during seasonal transitions (e.g. winter gas demand) or when a short-term shock affects only part of the maturity spectrum.	+2 \$/bbl
τ	**Decay speed**	How quickly the curve converges to β_0. A small τ means the shock is short-lived; a large τ means the market expects conditions to persist for many months.	6 months

Robust fitting: A naive fit often produces degenerate results where one parameter diverges to an extreme value (e.g. $\beta_0 = 3671$ for Palladium at \$1 100/oz). CFCAP uses a three-attempt strategy: (1) full NS with β_2 bounded to $\pm 8\%$ of spot; (2) NS with $\beta_2 = 0$ for monotone curves; (3) exponential fallback $F(T) = F_\infty + (S - F_\infty)\, e^{-kT}$ for sparse curves with fewer than 5 contracts.

4.2 Implied Convenience Yield

The convenience yield is one of the most important quantities in commodity finance. It measures the implied benefit of holding the *physical* commodity rather than a futures contract — the insurance value of immediate availability.

It is derived by inverting the cost-of-carry equation:

$$F(T) = S \cdot e^{(r+u-cy)\,T} \quad \Longrightarrow \quad cy(T) = r + u - \frac{1}{T}\ln\left(\frac{F(T)}{S}\right) \tag{2}$$

where r is the risk-free rate, u the annualised storage cost, and S the spot price.

Condition	Signal	Economic interpretation
$cy > 2 \times r$	**BUY**	The market is paying a large premium for physical ownership. Inventories are likely low, supply is tight, and holders of the physical commodity earn a substantial implied yield.
$r < cy < 2 \times r$	Neutral	Positive carry, but not strong enough to drive aggressive accumulation.
$0 < cy < r$	Neutral	Positive but below the cost of financing. Physical storage is barely economic.
$cy < 0$	**SELL**	The forward premium exceeds all carry costs. A classic cash-and-carry arbitrage: buy spot, store, sell forward at a profit. Signals a surplus market.

4.3 Roll Yield

When an investor holds a long futures position, the front-month contract expires every month and must be *rolled* into the next one. The cost or gain of this operation is the roll yield:

$$ry(T) = \frac{S - F(T)}{F(T) \cdot T} \tag{3}$$

In backwardation ($S > F(T)$), the roll yield is **positive**: rolling from M1 to M2 generates a gain because M2 is cheaper than M1, and it will eventually converge up to the spot price. This rewards passive long investors.

In contango ($S < F(T)$), the roll yield is **negative**: each monthly roll costs money because the new contract is more expensive. This drag can amount to 10–15% per year and explains why many commodity ETFs significantly underperform the spot price over time.

4.4 Calendar Spreads

A calendar spread is the price difference between two futures contracts:

$$\text{Spread}_{i \to j} = F(T_j) - F(T_i), \quad T_j > T_i$$

CFCAP computes all consecutive M+1 spreads along the curve. A negative spread means the near contract is more expensive (backwardation); a positive spread means the deferred contract is more expensive (contango).

The platform counts how many months are in each regime and highlights the most significant spread legs for potential calendar spread trades.

5 Trading Signal Engine

5.1 Overview

The signal engine is one of the most distinctive features of CFCAP. Rather than simply displaying numbers, it evaluates **51 rule-based conditions** against the live analytics and produces actionable outputs with a plain-language rationale for each one.

Every signal card in the dashboard shows:

- The **signal level**: BUY, SELL, SPREAD, HEDGE, ARBITRAGE, WARNING, or NEUTRAL
- The **trigger values**: the exact numbers that activated the rule (e.g. `CY +14.2% » 2.5x RF (12.5%)`)
- The **rationale**: a short explanation of why the condition is significant and what it implies for positioning

5.2 Signal Categories

Category	Signals	What it detects
Convenience Yield	6	$cy > 2.5r$: physical accumulation. $cy < 0$: cash-and-carry arbitrage. CY term structure steepening or flattening.
Roll Yield	6	$ry > 8\%$/yr: rewards long exposure. $ry < -8\%$/yr: passive long drag. Net carry cost vs. convenience yield.
Calendar Spreads	8	M1-M2, M1-M3, M1-M6 spreads against spot thresholds. Mixed structure (front backw. + back contango). Butterfly spreads.
Nelson-Siegel	6	Spot vs. β_0 fair value (mean reversion). β_2 hump or valley. Fast vs. slow τ decay signal.
Structural Regime	5	Proportion of curve in backwardation or contango, from mild ($\geq 60\%$) to deep ($\geq 85\%$). Transitional market.
Temporal (J-7/J-14)	6	Price momentum over 7 days. Curve twist (front vs. back shift). Parallel shift detection. Price acceleration.
Hedger Signals	3	Producer sell-forward opportunity. Consumer fixed-price buying. Collar strategy for balanced markets.
Arbitrage	2	Actual forward vs. theoretical $S \cdot e^{(r+u)T}$. Reverse cash-and-carry when $cy \gg r + u$.
Curve Shape	3	$R^2 < 0.75$: kink or non-linear structure. High NS residuals. Implied volatility from curve steepness.
Risk	2	$\sigma_{\text{ann}} = \frac{\lvert\Delta S_{7d}\rvert}{7}\sqrt{252}$. VaR at 95% and 99% from 7-day move.
Summary	2	≥ 3 aligned signals $\Rightarrow$ **STRONG BUY** or **STRONG SELL**.
Total	**51**	

Example: On a WTI curve in deep backwardation with $cy = 18\%$ and $r = 4.25\%$, the engine would simultaneously generate: a **BUY** on convenience yield ($cy > 2.5 \times r = 10.6\%$), a **BUY** on roll yield ($ry > 8\%$/yr), a **BUY** on structural regime ($\geq 85\%$ months in backwardation), a **HEDGE** for producers (lock in current elevated prices), and a **STRONG BUY** summary (4 aligned signals). Each card explains the numbers and the reasoning in plain language.

6 EIA Fundamental Data

For energy commodities, the shape of the forward curve is ultimately driven by physical fundamentals: how much crude oil is sitting in storage tanks, how much is being produced, and how quickly refineries are processing it. CFCAP integrates this context directly through the **EIA Open Data API**.

Series	Unit	Why it matters for the forward curve
Cushing crude stocks	kbbl	The key WTI delivery point. Falling stocks support backwardation; rising stocks drive contango.
US total crude stocks	kbbl	Broad supply picture. A consecutive build of 3+ weeks is a bearish signal.
US crude production	kbbl/d	Higher production ⇒ more supply pressure, supports contango.
WTI / Brent spot	$/bbl	Cross-check against downloaded futures to detect data errors.
US refinery utilisation	%	Low utilisation ⇒ crude accumulates. High utilisation ⇒ demand for crude supports backwardation.
US natural gas storage	Bcf	Below the 5-year average ⇒ bullish signal for TTF and Henry Hub.

The EIA data is cached locally for 24 hours to avoid redundant API calls. Each inventory figure is displayed with its week-on-week change and direction, alongside a 52-week time series chart.

Practical example: If Cushing stocks have fallen for four consecutive weeks and the WTI curve is in strong backwardation, the EIA panel and the signal engine will both point in the same direction — reinforcing confidence in a bullish view. If they diverge (e.g. backwardation but rising stocks), it is a signal to investigate further before acting.

7 Historical Comparison and Persistence

A single forward curve snapshot is informative, but the real insight comes from comparison over time. CFCAP saves a CSV snapshot of every curve it downloads, organised by commodity and date.

```
data/
|-- curves/wti_crude_oil/
|    |-- 2026-04-07.csv     # today
|    |-- 2026-03-31.csv     # 7 days ago
|    |-- 2026-03-24.csv     # 14 days ago
|    |-- ...
```

```
|-- dashboards/wti_crude_oil/
|   |-- wti_crude_oil_20260407_091522.png
|-- eia_cache/            # 24h TTL
|-- logs/scheduler.log
```

After two weeks of daily scheduler runs, the J-7 and J-14 comparisons on the Forward Curve chart use **real historical data** rather than simulated shifts. This allows the user to see exactly how the curve has moved and whether the move was a parallel shift (macro driver) or a twist (supply shock concentrated at the front end).

The Streamlit dashboard includes a **date selector** in the Forward Curve tab. The user can overlay the current curve with any saved snapshot to visualise the exact curve shape before and after any market event.

8 Potential Applications

CFCAP is designed to be useful to a wide range of users in the commodity and energy sectors.

User	How CFCAP helps them
Energy trader	Identify carry trade opportunities, detect prompt supply tightening before it is widely reported, determine the optimal maturity to roll futures positions, and monitor structural regime changes across multiple commodities simultaneously.
Risk manager (airline, refiner, utility)	Monitor the roll cost of existing hedging programmes, time forward purchases when the curve structure favours buyers, evaluate collar strategies in balanced markets, and receive hedger signals tailored to each commodity.
Portfolio manager	Assess whether passive long exposure is rewarded by the current roll yield, screen multiple commodity markets for carry opportunities, and track structural regime changes across an entire commodity portfolio.
Analyst / researcher	Study how forward curves evolve across commodity cycles, compare seasonal patterns between years, test the relationship between EIA fundamental data and curve shape, and build historical datasets automatically via the scheduler.
Student / practitioner	Apply cost-of-carry theory, Nelson-Siegel fitting, implied yields and term structure analysis to live market data, with all calculations transparent and well-documented.

9 Possible Improvements

The platform is already functional across 65 commodities. Several directions could extend its analytical depth and usability.

9.1 Quantitative Extensions

Nelson-Siegel-Svensson model. The NSS model adds a second curvature term, making it better suited to longer-dated curves with more complex shapes such as TTF Natural Gas at 24 months:

$$F(T) = \beta_0 + \beta_1\,\varphi(T,\tau_1) + \beta_2\big[\varphi(T,\tau_1) - e^{-T/\tau_1}\big] + \beta_3\big[\varphi(T,\tau_2) - e^{-T/\tau_2}\big]$$

Asian option pricer. Many commodity hedges use options on the *average* price over a period rather than a fixed-date option. Using the NS term structure as the diffusion input and a Monte Carlo simulation, CFCAP could estimate the fair value of Asian calls and puts — directly applicable to jet fuel and crude oil hedging programmes.

Historical realised volatility. Once a sufficient number of daily snapshots have been saved, rolling 20-day and 60-day realised volatility can be computed from the daily spot returns. A volatility cone overlaid on the forward curve would help option traders assess whether implied volatility is rich or cheap.

Portfolio VaR. By defining a book of futures positions (volume, maturity, direction) and applying Monte Carlo curve shocks, the platform could compute Value-at-Risk at 95% and 99% confidence for a realistic commodity derivatives portfolio.

9.2 Data and Infrastructure

SQLite database. Replacing the CSV files with a local SQLite database would allow faster historical queries, rolling cross-commodity correlation analysis, and more complex multi-date comparisons.

FastAPI REST endpoint. Exposing the analytics as JSON at `/curve/{commodity}` and `/signals/{commodity}` would make the platform consumable by Excel via Power Query, Power BI dashboards, or other automated scripts.

Configuration file. Moving API keys, the scheduler time and the default batch list to a `config.yaml` file (excluded from version control) would follow security best practices and make the platform easier to deploy.

9.3 User Experience

Morning report. An automated HTML report generated after each scheduler run, containing all charts and signal summaries, could be sent by email every morning before the trading day begins.

Real-time alerts. Email or Telegram notifications triggered by specific conditions — such as a convenience yield crossing $2 \times r$ or a regime shift from contango to backwardation — would add a real-time monitoring layer.

Multi-commodity overview. A landing page in Streamlit showing one row per commodity, with spot, structure, roll yield and dominant signal, would allow a trader to monitor an entire portfolio at a glance.

10 Conclusion

CFCAP demonstrates how a complete, professional-grade quantitative analytics platform can be built with Python and freely available market data.

By combining live data acquisition from two sources, Nelson-Siegel curve fitting with a robust multi-attempt strategy, cost-of-carry analytics including convenience yield and roll yield, a 51-signal trading engine grounded in financial theory, and an interactive browser dashboard, the platform provides a rigorous and practical tool for anyone working with commodity derivatives.

Its single-file architecture makes it easy to deploy and extend: new commodities can be added to the registry in a few lines, new signals can be plugged into the engine without touching the rest of the code, and the modular design makes it straightforward to integrate more advanced models as the project evolves.